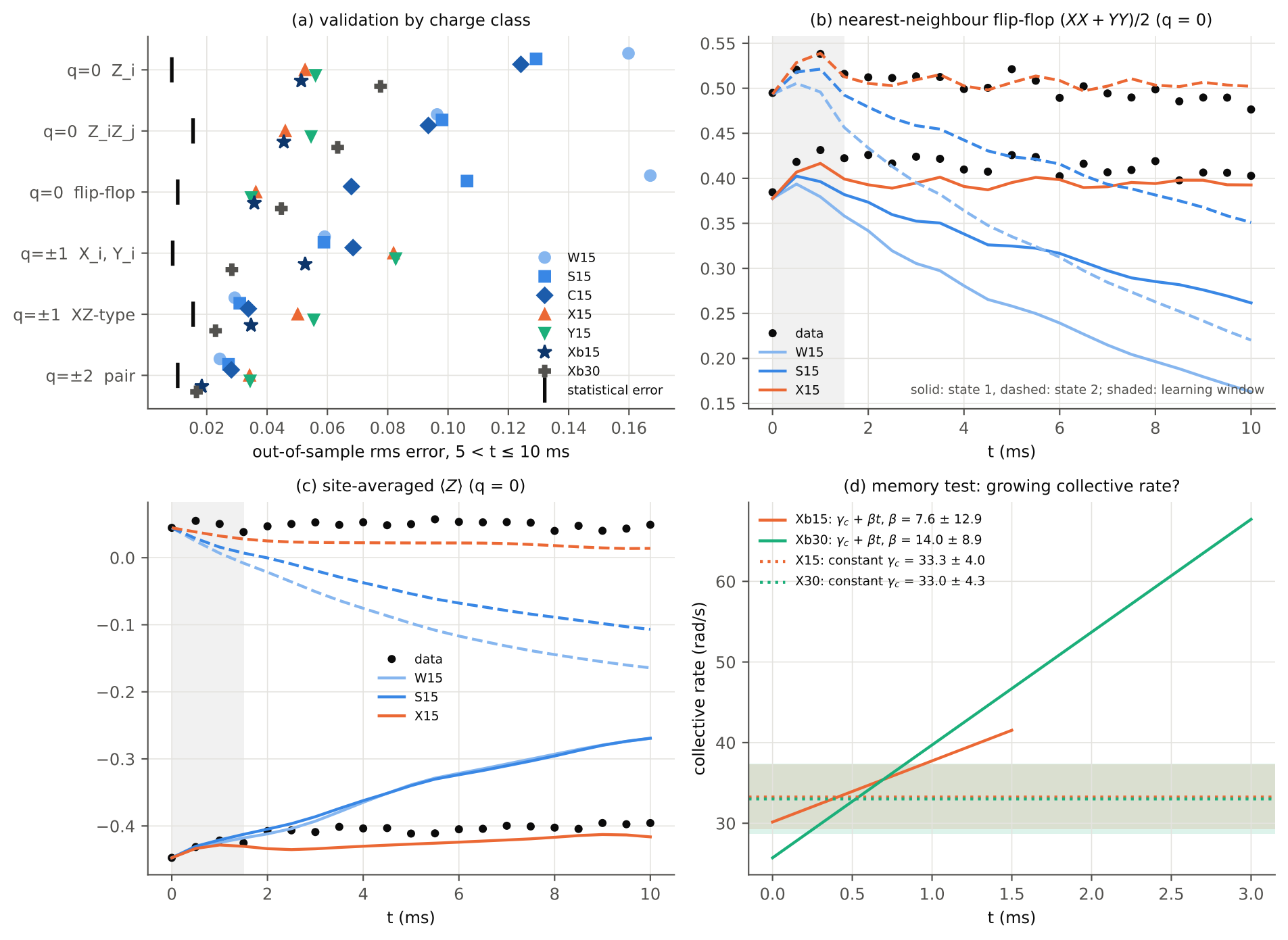
(a) validation by charge class
q=0 Z_i
q=0 Z_iZ_j
q=0 flip-flop
q=±1 X_i, Y_i
q=±1 XZ-type
q=±2 pair
W15
S15
C15
X15
Y15
Xb15
Xb30
statistical error
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16
out-of-sample rms error, 5 < t ≤ 10 ms
(b) nearest-neighbour flip-flop (XX + YY)/2 (q = 0)
0.55
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0.15
data
W15
S15
X15
solid: state 1, dashed: state 2; shaded: learning window
0
2
4
6
8
10
t (ms)
(c) site-averaged ⟨Z⟩ (q = 0)
0.0
−0.1
−0.2
−0.3
−0.4
data
W15
S15
X15
0
2
4
6
8
10
t (ms)
(d) memory test: growing collective rate?
Xb15: $\gamma_c + \beta t$, β = 7.6 ± 12.9
Xb30: $\gamma_c + \beta t$, β = 14.0 ± 8.9
X15: constant γ_c = 33.3 ± 4.0
X30: constant γ_c = 33.0 ± 4.3
collective rate (rad/s)
60
50
40
30
0.0
0.5
1.0
1.5
2.0
2.5
3.0
t (ms)